

DCMG Controller Implementation Review Checklist

Prepared from the theory-to-code review of P6_Data_Driven_Co_Design.pdf and the current MATLAB implementation. This document combines the controller-design, DG/line-model, simulation, data-generation, and main-workflow findings.

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How to use this document

- [] means a decision or change remains to be made.
- [x] means the implementation was checked and matches the stated paper model at the reviewed level.
- **P1** should be resolved before claiming the theoretical certificate is implemented.
- **P2** can materially change trajectories or the model/data-driven comparison.
- **P3** concerns numerical robustness, reproducibility, diagnostics, or maintenance.
- Search the MATLAB files for % REVIEW PROPOSAL <ID> to find disabled code suggestions.
- ADD proposals can normally be uncommented directly.
- For REPLACE proposals, disable the identified active line/block before uncommenting the replacement.
- DESIGN and CHECK proposals require a modeling choice rather than a blind code edit.

No proposal has been enabled. The baseline simulation behavior is unchanged.

1. Prioritized revision queue

Use this section as the implementation order. The later sections retain the same revisions grouped by subsystem and provide the detailed rationale.

Effort estimates:

- **E1 - Quick:** localized guard, assertion, option, or diagnostic; normally under one hour including a focused test.
- **E2 - Moderate:** coordinated edits across a few methods or a new regression experiment; normally several hours.
- **E3 - Deep:** requires a theoretical/modeling decision, experiment redesign, retuning, and broad revalidation.

The groups are ordered by a combined criticality-and-ease score; ties favor criticality. Items inside each group are also in recommended execution order. Some E1 changes may expose infeasibility. That is a useful result, not a reason to restore an invalid certificate.

A. P1 + E1 - critical quick wins

- [] Abort MB/DD global design immediately when any local design fails; reject nonfinite or incorrectly signed ν_i . See MB-G01 and DD-G01.
- [] Move every `value(...)`, minor check, and gain recovery after `sol.problem==0`.
- [] Assert recovered $Y_{22}<0$, $X_{i11}>0$, $X_{i22}<0$, and finite/nonzero local dissipativity parameters.
- [] Assert the recovered model-based local closed loop is Hurwitz with a numerical margin. See MB-L03.
- [] Restore the hard, unsoftened MB and DD global LMIs for certificate-producing runs. See MB-G03 and DD-G05.
- [] Report and require the minimum eigenvalue of each unsoftened certificate; enforce $\gamma_{sq}>0$, $\lambda_{\min}>0$, and $Q>0$.
- [] Remove the redundant $1e6$ multiplier from the DD physical equality. See DD-G06.
- [] Decouple DD ϵ from γ_{sq} ; the current bounds force both to $1e-3$. See DD-G04.
- [] Stop perturbing the stabilizing GSC gain by up to 200% during data generation. See GSC-01.
- [] Disable independent network-current clipping in the certified linear run and retain it only as a separately labeled stress mode. See DG-03 and DATA-05.
- [] Disable or explicitly label VSC command saturation as outside the linear certificate. See DG-04 and DATA-06.
- [] Assert equilibrium dynamics and line-current residuals after every successful steady-state solve. See SS-03.

B. P1 + E2 - critical moderate revisions

- [] Check $\text{rank}([U_i; X_i])$ and its smallest singular value before creating DD YALMIP variables; include the uncontrollable line-current row. See DATA-01 and DD-L01.
- [] Use the exact per-DG Q_{Bar_w} normalization again when assembling Proposition 8. See DD-L02 and DD-G02.

- [] Use one documented T_s and hold every input component over each identification interval, including Stages 1 and 5.
- [] Resolve whether σ_{noise} is a covariance or a factor and make generation, logging, and documentation consistent. See DATA-07, DATA-08, and MAIN-01.
- [] Add Eq. (45) bounds on equilibrium line current and VSC command, and replace $+\epsilon I$ with a stated sharing target. See SS-01 and SS-02.
- [] Verify the Q_{Bar}_w sign convention and all required positivity conditions after scaling.
- [] Confirm that output data meet the paper's noise-free-output assumption or explicitly adopt the augmented measurement-noise model.

C. P2 + E1 - important quick wins

- [] Handle an all-zero global gain and recheck stability after communication-gain thresholding. See GSC-02 and GSC-03.
- [] Use exact-ZOH matrices for DD diagnostic poles and require the recovered sampled loop to be Schur stable. See DD-L05 and DD-L06.
- [] Re-enable the physical consistency residual $D' * K - P * Y_{\text{Bar}} * D'$ and fail when it exceeds tolerance.
- [] Replace determinant/leading-minor diagnostics with minimum eigenvalues. See NUM-01 and NUM-02.
- [] Use $\text{norm}(\text{diag}(\epsilon_{\text{slack}}), 1)$ when an L1 sum of diagonal slacks is intended.
- [] Correct event times for nonzero $t_{\text{span}}(1)$. See SIM-01.
- [] Make the state-disturbance draws symmetric in both states. See SIM-02 and SIM-05.
- [] Decide and encode whether heightened noise is a short pulse or a full-stage interval. See SIM-03 and SIM-06.
- [] Initialize comps, deduplicate proximity/MST edges, and handle parallel lines consistently. See MAIN-04, MAIN-05, and SYS-02.
- [] Guard line-current per-unit division by zero or a near-zero equilibrium value. See MAIN-07.
- [] Create Results/ before saving and assert compatible horizons before comparison. See MAIN-08 and MAIN-10.
- [] Apply axis limits to explicit axes and make $\text{adjustAxesPadding}()$ match its documented fractional units. See MAIN-09 and MAIN-11.
- [] Add missing-noise guards, physical-parameter validation, endpoint validation, and coincident-position protection. See DG-01, DG-02, LINE-01, and DRAW-01.
- [] Remove the inactive DD BMI warning and guard dimension-dependent debug slices.
- [] Correct the ϵI numeric format and resolve the documented $I_{\text{rated}} = 2 * P_r / V_r$ factor. See MAIN-02.

D. P1 + E3 - critical design/theory work

- [] Define one theorem-consistent sampled experiment satisfying $X_{\text{bar}} = A_d X + B_d U + W$ exactly.
- [] Specify an a-priori Q_w independently of the realized data and unknown A/B, with a justified reserve margin.
- [] Remove true-model A/BBar from DD synthesis inputs; use logged disturbance information with a consistent exact-discrete interpretation.
- [] Verify matrix S-lemma regularity/strict feasibility for every local QMI and the aggregate QMI.
- [] Decide whether Proposition 5 remains relaxed or Remark 10 is enforced; if enforced, iterate the BMI surrogate until the equality residual is acceptable. See MB-L01.

E. P2 + E2 - important moderate revisions

- [] Add a bounded persistently exciting voltage-command signal instead of relying on initial transients/noise.
- [] Nondimensionalize states and inputs, then define dimensionless communication budgets and gain bounds shared by MB and DD.
- [] Correct local/global objective directions for ϵ , $\text{trace}(P)$, γ_{Sq} , and λ ; add required upper bounds before maximizing margins.
- [] Use the same communication-cost budget for MB and DD before comparing performance. See DD-G03.
- [] Perturb physical L, C, R_f , and RL and rebuild the model instead of perturbing arbitrary entries of A. See SIM-04.
- [] Derive stage times and solver output spacing from inputs rather than hard-coding $dt = 1e-5$.
- [] Add usable steady-state weights/bounds through opts and explicit converter command/current ratings.
- [] Prevent public line R and cached g from diverging. See LINE-02.
- [] Decide the active load ripple, initial-state range, bounded α_Z , and converter-rating interpretation used in reported experiments. See DG-05, MAIN-03, and MAIN-06.
- [] Integrate logged process disturbance over each sample interval consistently rather than blindly multiplying by T_s .

- [] Store local/global gains, minimum eigenvalues, solver status, physical residuals, rank diagnostics, and spectral radii in returned out structs.

F. P3 + E1 - robustness and documentation quick wins

- [] Record MATLAB/YALMIP/MOSEK versions, seed, noise interpretation, disturbance bound, and full DG/line parameters with every run.
- [] Correct or remove unused integral-action dimensions and fields (z , K_I).
- [] Remove clearly unused variables only after the final theory form is selected.
- [] Store unpruned and pruned gains separately with topology, cost, and recertification results.

G. P2 + E3 - important experiment/model redesign

- [] If limiters are part of the claimed model, replace clipping with a physical limiter model and extend the analysis; otherwise keep limiter tests separate.
- [] Define structured component uncertainty, regenerate DD data after the event, and keep uncertainty experiments separate from nominal validation.
- [] Run a fair MB/DD comparison using the same plant, normalization, budgets, disturbance schedule, and communication constraints.

H. P3 + E2/E3 - final reproducibility and paper pass

- [] Complete the controlled experiment protocol in Section 12 after Groups A-G are resolved.
- [] Store the minimum result record in Section 13 for every accepted run.
- [] Generate paper plots only from runs whose configuration and certificate diagnostics are stored with the trajectories.

2. Confirmed theory-to-code matches

- [x] `DG.updateModel()` implements paper Eq. (39) with state $x_i = [V_i; I_{ti}]$:
 - $A_i = [-1/(R_L C), 1/C; -1/L, -R_f/L]$.
 - $B_i = B_{wi} = [-1/C, 0; 0, 1/L]$ through $BBar = [E, B]$.
 - $\theta_i = E \cdot I_{bar}$ has the correct constant-current-load sign.
- [x] `TransmissionLine.current()` implements $(V_i - V_j)/R_{ij}$.
- [x] `DCMicrogrid.dynamics()` computes exported network current as $YBar \cdot V$, consistent with Eq. (41).
- [x] The local controller controls only the VSC voltage-command row; the uncontrollable line-current row is fixed to zero in local-gain decision variables.
- [x] `designLocalXiDissipative()` contains the continuous-time local LMI corresponding to Eq. (24) with $C_i = I$.
- [x] Its second local LMI follows the relaxed necessary condition in Eq. (21).
- [x] `designLocalXiDissipative_DataDriven()` follows the main block structure of Proposition 7, Eqs. (33)-(34).
- [x] Data-driven local gain recovery $K_i = K_{ilde_i} \cdot (X_i \cdot \hat{K}_{hat_i})^{(-1)}$ is implemented as right division.
- [x] `codesign_MB_DRC()` structurally enforces the physical line-current interconnection in the full pre-scaled gain.
- [x] `codesign_DD_DRC()` follows the major Q, S, and R block construction of Proposition 8, Eqs. (35)-(36).
- [x] `solveSteadyState()` enforces the equilibrium physics and proportional internal-current sharing in Eqs. (44), (46), and (47).

3. Shared theoretical assumptions

Assumptions 1-3

- [] **P1:** Explicitly assert $Y_{22} < 0$ and $X_{i11} > 0$, $X_{i22} < 0$ after recovery, even when principal LMI blocks imply them numerically.
- [] **P1:** Confirm that the scalar-inner-block parameterization required by Assumption 3 is the intended restriction for every experiment.
- [] **P1:** Abort the global design if any local design fails. Current code warns and continues, which can lead to division by $\nu_i = 0$. See MB-G01 and DD-G01.
- [] **P1:** Check every recovered ν_i is finite, nonzero, and has the sign required by $X_{i11} = -\nu_i \cdot I > 0$.

Assumptions 4-5

- [] **P1:** Supply an a-priori quadratic disturbance bound satisfying Eq. (26). Do not infer the only bound from the same realized disturbance trajectory used for design.

- [] **P1:** Check the matrix S-lemma regularity/strict-feasibility condition for every local QMI and the aggregate QMI.
- [] **P1:** Verify $\text{rank}([U_i; X_i]) = n_{ui} + n_{xi}$ and report the smallest singular value for each DG. See DATA-01 and DD-L01.
- [] **P1:** Confirm output data are noise-free as assumed by the paper. Process noise is allowed; measurement noise needs the augmented uncertainty model discussed in Remark 13.
- [] **P2:** Add a persistently exciting, bounded additive voltage-command signal during data collection instead of relying on initial transients and process noise.
- [] **P2:** Confirm the uncontrollable line-current component still provides sufficient independent data for the four-row $[U_i; X_i]$ rank test.

4. Model-based local design

Function: `DG.designLocalXiDissipative()`

- [] **P1:** Decide whether to use the relaxed Proposition-5 condition or enforce the optional Remark-10 equality.
- [] **P1:** If enforcing Remark 10, iterate `xBar11ValGuess` until $x_{\text{Bar}11} * K_{\text{Hat}} = x_{21} * K_{\text{Tilde}}$ is satisfied. A warning alone does not restore the equality. See MB-L01.
- [] **P1:** Move all `value(...)`, `minor`, and `gain recovery` operations after checking `sol.problem==0`; failed solves currently produce meaningless post-processing.
- [] **P1:** Verify the recovered continuous closed loop is Hurwitz with a numerical margin. See MB-L03.
- [] **P2:** Revisit the objective. Minimizing `+epsilon` drives the strictness margin to its lower bound. See MB-L02.
- [] **P2:** Nondimensionalize voltage/current states and both input channels before choosing objective weights.
- [] **P3:** Remove unused variables (`YBar`, unused output selector variables) after the theory form is finalized.
- [] **P3:** Store `minEig(LMI1)`, `minEig(LMI2)`, BMI residual, gain norm, and solver residuals in `out`.

5. Data-driven local design

Function: `DG.designLocalXiDissipative_DataDriven()`

- [] **P1:** Make data satisfy one discrete equation $X_{\text{bar}} = A_d X + B_d U + W$ exactly for the selected interpretation.
- [] **P1:** Hold all input components over $[kT_s, (k+1)T_s)$ during the identification/design trajectory.
- [] **P1:** Do not compute the required disturbance description from true A and B_{Bar} in a direct data-driven claim. See DATA-02 and DATA-03.
- [] **P1:** Use the exact per-DG $Q_{\text{Bar_w}}$ scaling again in Proposition 8. See DD-L02 and DD-G02.
- [] **P1:** Check Assumption 5 before creating YALMIP variables. See DD-L01.
- [] **P1:** Confirm the paper's $Q_{\text{Bar_w}}$ sign convention and strict S-lemma regularity numerically after scaling.
- [] **P2:** Replace the $1e-9$ feasibility margin with a tolerance meaningful at normalized scale. See DD-L03.
- [] **P2:** Rework the objective so it does not minimize the margin. See DD-L04.
- [] **P2:** Use exact ZOH matrices for diagnostic eigenvalues, not forward Euler, when validating the ode45/ZOH implementation. See DD-L05.
- [] **P2:** Require the recovered sampled closed loop to be Schur stable. See DD-L06.
- [] **P2:** Remove the unused `xBar11ValGuess` warning while its BMI constraint is disabled; it currently suggests a consistency requirement that is not imposed.
- [] **P3:** Remove debugging slices such as `Q_w(1:5, 1:5)` or guard them by dimensions.
- [] **P3:** Store rank, singular values, `lambda1`, `lambda2`, LMI minimum eigenvalues, and spectral radius in `out`.

6. Model-based global co-design

Function: `DCMicrogrid.codesign_MB_DRC()`

- [] **P1:** Abort immediately if any local Proposition-5 problem fails. See MB-G01.
- [] **P1:** Remove the negative soft diagonal slack from certificate-producing runs. $W - \epsilon_{\text{Slack}} \geq \epsilon I$ is easier than $W \geq 0$ when $\epsilon_{\text{Slack}} < 0$. See MB-G03.
- [] **P1:** Only label a result Proposition-4 feasible if the unsoftened W has nonnegative minimum eigenvalue within solver tolerance.
- [] **P1:** Enforce/verify $\gamma_{\text{Sq}} > 0$ when it represents Y_{11} .
- [] **P2:** Optimize γ_{Sq} if figures or text claim an optimized L2 gain; its objective coefficient is currently zero.

- [] **P2:** Reconsider $\text{+trace}(P)$. Remark 6 proposes $-\phi(p)$ to encourage larger p_i ; minimizing trace has the opposite direction. Add $P \leq p_{\text{Max}} I$ before using a negative trace term. See MB-G02 and MB-G04.
- [] **P2:** Use $\text{norm}(\text{diag}(\text{epsilonSlack}), 1)$ if an L1 sum is intended; $\text{norm}(\text{epsilonSlack}, 1)$ is an induced matrix norm.
- [] **P2:** Confirm the communication-cost budget and gain bounds are dimensionless after state/input scaling.
- [] **P3:** Re-enable and assert the physical consistency residual $D' * K - P * Y_{\text{Bar}} * D'$.
- [] **P3:** Store the unpruned gain, pruned gain, certificate eigenvalue, physical residual, and communication cost separately.

7. Data-driven global co-design

Function: `DCMicrogrid.codesign_DD_DRC()`

- [] **P1:** Abort if any local Proposition-7 design fails. See DD-G01.
- [] **P1:** Assemble aggregate Q_{Bar}_w from the same normalized blocks used locally. See DD-G02.
- [] **P1:** Restore the hard Proposition-8 LMI for certificate-producing runs. See DD-G05.
- [] **P1:** Remove the $1e6$ multiplier on the physical equality. It is algebraically redundant and can damage solver scaling. See DD-G06.
- [] **P1:** Decouple ϵ from γ_{Sq} . Current bounds force both to exactly $1e-3$. See DD-G04.
- [] **P1:** Verify $Q > 0$, $\lambda_{\text{bda}} \geq 0$, and the minimum eigenvalue of the unsoftened final LMI.
- [] **P2:** Use the same dimensionless communication budget as the model-based design before comparing performance. Current values are 1 and $1e-3$. See DD-G03.
- [] **P2:** Align the objective with Proposition 8/Remark 6; current $\text{+trace}(P)$ and +epsilon directions are questionable. See DD-G07.
- [] **P2:** Either penalize λ as documented or remove the claim that the objective makes it small.
- [] **P2:** Replace determinant-based diagnostics with minimum eigenvalues. See NUM-01.
- [] **P3:** Report how QMI normalization changes the recovered λ ; do not compare raw λ values across differently scaled runs.
- [] **P3:** Validate the recovered gain against the known simulated model as a diagnostic only, clearly separated from the data-driven synthesis inputs.

8. Model-based stabilizing baseline

Function: `DCMicrogrid.design_MB_GSC()`

- [] **P1:** Do not perturb a stabilizing gain by up to 200% to generate data without rechecking stability. Prefer bounded additive PE input. See GSC-01.
- [] **P2:** Confirm the dense full gain satisfies the physical line-current equality before extracting the VSC command rows.
- [] **P2:** Rework the objective if ϵ is intended to measure robustness; $\epsilon + \text{trace}(P)$ minimizes ϵ .
- [] **P2:** Handle the all-zero gain case in communication thresholding. See GSC-02.
- [] **P2:** Recheck stability after thresholding small communication blocks. See GSC-03.

9. Equilibrium design

Function: `DCMicrogrid.solveSteadyState()`

- [] **P1:** Add Eq. (45) bounds on equilibrium exported current and VSC command. See SS-01.
- [] **P1:** Replace +epsI , which drives current utilization toward zero, with a stated target such as $(\text{epsI} - 1)^2$ or a justified alternative. See SS-02.
- [] **P1:** Assert equilibrium and line-current residuals after a successful solve. See SS-03.
- [] **P2:** Add optional weights/bounds through the currently unused `opts` input.
- [] **P2:** Add explicit converter command/current ratings rather than relying on later simulation clipping.
- [] **P3:** Correct the z/K_I documented dimensions or remove unused integral-action fields.
- [] **P3:** Use a numeric format such as `% .4f` when printing epsI , rather than `%s`.

10. DG and transmission-line runtime model

- [] **P1:** Remove independent line-current clipping for the linear/theory-consistent run. It can violate KCL because each DG clips a current already determined by $Y_{\text{Bar}} * V$. See DG-03 and DATA-05.

- [] **P1:** Treat VSC command saturation as an uncertified nonlinear stress test, or include a limiter model in the analysis. See DG-04 and DATA-06.
- [] **P2:** Guard missing noise data for deterministic unit tests. See DG-02.
- [] **P2:** Decide whether `iload_fun` is active; it is configured but unused. See DG-05.
- [] **P2:** Validate positive L, C, RL, line R, and valid endpoints. See DG-01 and LINE-01.
- [] **P2:** Prevent public R and cached g from diverging. See LINE-02.
- [] **P2:** Accumulate or prohibit parallel lines consistently. See SYS-02 and MAIN-05.
- [x] **DG.**`draw()` and `TransmissionLine.draw()` now attach all text to the caller-provided axes and retain their original call forms.
- [] **P3:** Guard coincident DG positions in communication-topology drawing. See DRAW-01.

11. Simulation and main workflow

Staged simulation

- [] **P1:** Use ZOH data collection in Stages 1 and 5 when the design interprets samples as discrete-time inputs.
- [] **P1:** Use a single documented T_s shared by data collection, design, ZOH simulation, and validation.
- [] **P2:** Correct event times for nonzero `tspan(1)`. See SIM-01.
- [] **P2:** Correct the state-disturbance expression; it currently only decreases voltage and never perturbs current. See SIM-02 and SIM-05.
- [] **P2:** Decide whether heightened noise is a 2% pulse or a full-stage interval. See SIM-03 and SIM-06.
- [] **P2:** Perturb physical L, C, RF, and RL, then call `updateModel()`, instead of perturbing every A entry. See SIM-04.
- [] **P2:** Derive stage times and solver output spacing from inputs rather than hard-coding `dt=1e-5` inside `simulate()`.
- [] **P3:** Add defaults/validation for `linkFiltThresh`, `useData`, T_s , and `tspan`.

Noise and disturbance data

- [] **P1:** Resolve whether `sigma_noise` is a covariance or a factor. Current code multiplies `randn` directly by it. See DATA-07, DATA-08, and MAIN-01.
- [] **P1:** Do not fit Q_w tightly to one observed residual without an a-priori margin. See DATA-04.
- [] **P2:** If using logged process disturbance in simulation, integrate it over the sample interval consistently rather than blindly multiplying by T_s .
- [] **P3:** Record the random seed, covariance/factor, realized disturbance norm, and chosen bound in every stored result.

Main file and plots

- [x] `main.mlx` is the sole canonical workflow and contains the disabled MAIN-* review proposals directly.
- [x] The redundant root `main.m` was removed; the stale output-heavy historical export remains archived as `archive/legacy/main_legacy_export.m`.
- [] **P2:** Initialize comps before graph-growth attempts. See MAIN-04.
- [] **P2:** Deduplicate proximity and MST edges. See MAIN-05.
- [] **P2:** Decide whether $I_{rated}=2*Pr/V_r$ is intentional and correct the displayed rating comments. See MAIN-02.
- [] **P2:** Prefer bounded heterogeneous α_Z if random load composition is restored. See MAIN-03.
- [] **P2:** Decide whether the 50%-100% initial-state range is too severe for the intended comparison. See MAIN-06.
- [] **P2:** Do not divide line current by a zero/near-zero equilibrium current. See MAIN-07.
- [] **P3:** Create `Results/` before saving on a clean checkout. See MAIN-08.
- [] **P3:** Make `adjustAxesPadding()` implement its documented fractional units. See MAIN-09.
- [] **P3:** Verify all compared result horizons and event schedules agree. See MAIN-10.
- [] **P3:** Apply axis limits to explicit axes handles. See MAIN-11.

12. Controlled experiment protocol

Use this order so each result isolates one source of behavior.

1. [] Save the current baseline outputs and solver logs unchanged.

2. ☐ Run deterministic model checks with noise, saturation, pruning, and parameter events disabled.
3. ☐ Verify A, BBar, YBar, equilibrium residuals, and current-sign conventions numerically.
4. ☐ Run MB-GSC without gain corruption and verify the exact closed-loop eigenvalues before/after pruning.
5. ☐ Run the model-based local/global design with hard LMIs and record all minimum eigenvalues.
6. ☐ Generate a ZOH PE dataset with fixed Ts, known seed, and independently specified Q_w.
7. ☐ Check Assumption 5 singular values and S-lemma regularity for every DG.
8. ☐ Run data-driven local designs and verify gain recovery, LMI eigenvalues, and sampled-model spectral radii.
9. ☐ Run data-driven global design with hard LMI, consistent QMI scaling, and the same communication budget as MB-DRC.
10. ☐ Compare MB and DD in the nominal linear case before adding parameter changes.
11. ☐ Add structured component uncertainty and regenerate DD data after the event.
12. ☐ Add saturation/noise stress tests separately and label them as outside the linear certificate where applicable.
13. ☐ Generate paper plots only from runs whose configuration and certificate diagnostics were stored with the trajectories.

13. Minimum result record for the paper

For every method/run, store:

- ☐ MATLAB, YALMIP, and MOSEK versions.
- ☐ Random seed and complete DG/line parameters.
- ☐ Ts, ODE tolerances, noise interpretation, and disturbance bound.
- ☐ Data ranks and singular values for every DG.
- ☐ Local nu, rho, gain, LMI minimum eigenvalues, and solver status.
- ☐ Global unpruned/pruned gains, topology, cost, gammaSq, lambdas, and physical residual.
- ☐ Unsoftened certificate minimum eigenvalue.
- ☐ Equilibrium states/inputs and residuals.
- ☐ Saturation activation counts.
- ☐ Event definitions and structured parameter changes.
- ☐ Trajectories using a documented, nonzero-safe per-unit base.

14. File organization completed

- ☒ Key runtime files are documented in README.md.
- ☒ Temporary experiments and data snapshots were moved under archive/ and described in archive/README.md.
- ☒ Paper figures and TrajResults.mat remain under Results/ with a directory description.
- ☒ macOS .DS_Store files are ignored for future additions.
- ☒ Class headers describe the runtime/design role of each key file.

15. Final acceptance questions

- ☐ Can every theoretical assumption used in the paper be pointed to in code and to a stored numerical check?
- ☐ Does the data-driven design avoid using the true model except for explicitly labeled post-design diagnostics?
- ☐ Does the implemented sampled experiment satisfy the same data equation used by the theorem?
- ☐ Are reported certificates evaluated without negative feasibility slack and before any uncertified gain pruning?
- ☐ Are model-based and data-driven methods compared with the same plant, budgets, normalization, disturbances, and event schedule?
- ☐ Are nonlinear saturation and current-limiting effects either modeled or clearly separated from certified linear results?